Investigating the Relationship Between Retrotransposon Composition and Infectivity for *T. spiralis* and *T. nelsoni*

Group 5

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Introduction

Trichinella Species – An Introduction

- *Trichinella* are parasitic roundworms that cause:
- **Trichinosis (CDC, 2024)**
  - Usually caused from eating undercooked infected meat.
  - Larvae migrate into skeletal muscle, where they establish infection.
  - Adult worms from larvae in the infected meat mature in the small intestine, leading to digestive issues such as but not limited to nausea, diarrhea, abdominal pain, etc.
  - If left untreated, they can enter the bloodstream and invade the muscle tissue.
- The genus contains multiple species and they have differences in biological characteristics, host interactions, geographical distribution (Murrell *et al*. 2000).
- There are multiple *Trichinella* species that are categorized into encapsulated and non-encapsulated groups (Murrell *et al*. 2000).
  - The two groups split approximately 21 million years ago.

T. spiralis vs. *T. nelsoni*

- *T. spiralis* and *T. nelsoni* are sister species, meaning they are phylogenetically closely related (Bolas-Fernández *et al*. 2003).
- However, they have different recorded parasitic capabilities (Bolas-Fernández *et al*. 2003).
- This poses an interesting question: what specific differences in their genomes might impact their parasitic capabilities and how?

T. spiralis and *T. nelsoni* have differences despite them being evolutionarily closely related (Pozio and Zarlenga, 2013)

Characteristic	***T. spiralis***	***T. nelsoni***
Encapsulation	Encapsulated	Encapsulated
Geographic Range	Widespread	Mostly equatorial Africa
Host Associations	Domestic and wild mammals	Mostly wild mammals
Epidemiology	Domestic and wildlife transmission cycles	Mostly wildlife transmission cycle
Infectivity	High infectivity to rodents and swine	Low infectivity to rodents and swine, primarily associated with predators and scavengers.

What are Transposable Elements (TEs)

- **DNA transposons** cut themselves out and paste into a new spot (Feschotte *et al.*, 2007).
  - Copy number might stay the same or increase depending on mechanism
- **Retrotransposons** copy and then paste themselves into a new spot. It uses an RNA intermediate conducts reverse transcription to do this (Feschotte *et al.*, 2007).
  - Adds a new DNA copy
- TE insertions can cause mutations, gene duplications, and changes in gene regulation (4) (Feschotte *et al.*, 2007).
  - For instance, TE insertion into or near a gene can disrupt its function, TE sequences may contain promoters, enhancers or transcription factor binding sites that influence nearby genes, impacting gene regulation, etc.

What are TE superfamilies?

- TE superfamilies are groups of related TEs (Feschotte *et al.*, 2007):
  - Common evolutionary origin
  - Same mechanism of movement within the genome.
- Different TE superfamilies can exhibit different genomic behaviour, which might have different implications on the host interactions that the two *Trichinella* species display (Feschotte *et al.*, 2007).
- Thus, comparing TE communities in each species may help to discover differences (Feschotte *et al.*, 2007).

Our research question and hypothesis

- Are there differences in DNA transposon vs. retrotransposon composition, retrotransposon abundance or retrotransposon superfamily composition between the two *Trichinella* species?
- We hypothesize *T. Spiralis* will have a higher retrotransposon abundance and/or a greater proportion of lower diverged retrotransposon abundance compared with *T. nelsoni* due to its increased infectivity.

Methods

Import data: Read in *Trichinella spiralis and Trichinella nelsoni* RepeatMaster datasets to R from csv files. Used tidyverse, ggplot, and viridis packages in R.

Define TE categories: Determine DNA transposon and retrotransposon classes with Copilot prompt and filter datasets to include only these queries.

Counting copies: Count the number of copies of DNA transposons and retrotransposons for each species and the number of copies of each unique retrotransposon class, sorted by abundance.

Summarizing count data: Create a summary table to compare the ratios of TE categories for both species and a summary table to collate the retrotransposon class counts for both species.

Reshaping data: Convert the retrotransposon class counts summary table to long format, with one row per class per species

Plotting count data: Create a stacked bar chart to compare the ratios of DNA transposons vs. retrotransposons in both species. Create a bar plot of retrotransposon class composition between T. nelsoni and T. spiralis.

Filter and summarize data for select retrotransposons: For each species, filter for queries belonging to chosen retrotransposon classes based on Figure 2 and stack both data frames together.

Plotting repeat landscape: Create histograms of percent divergence with the two species overlaid with one panel per class.

Statistical tests: Performed Chi squared and Kolmogorov-Smirnov (KS) tests on the summarized datasets to determine statistical significance of the results

Figures

Figure 1. No statistical difference in the DNA transposons and retrotransposons proportions between *T. nelsoni* and *T. spiralis,* indicated by chi-squared test ($X^2 = 0.25$, df = 1, p = 0.62)

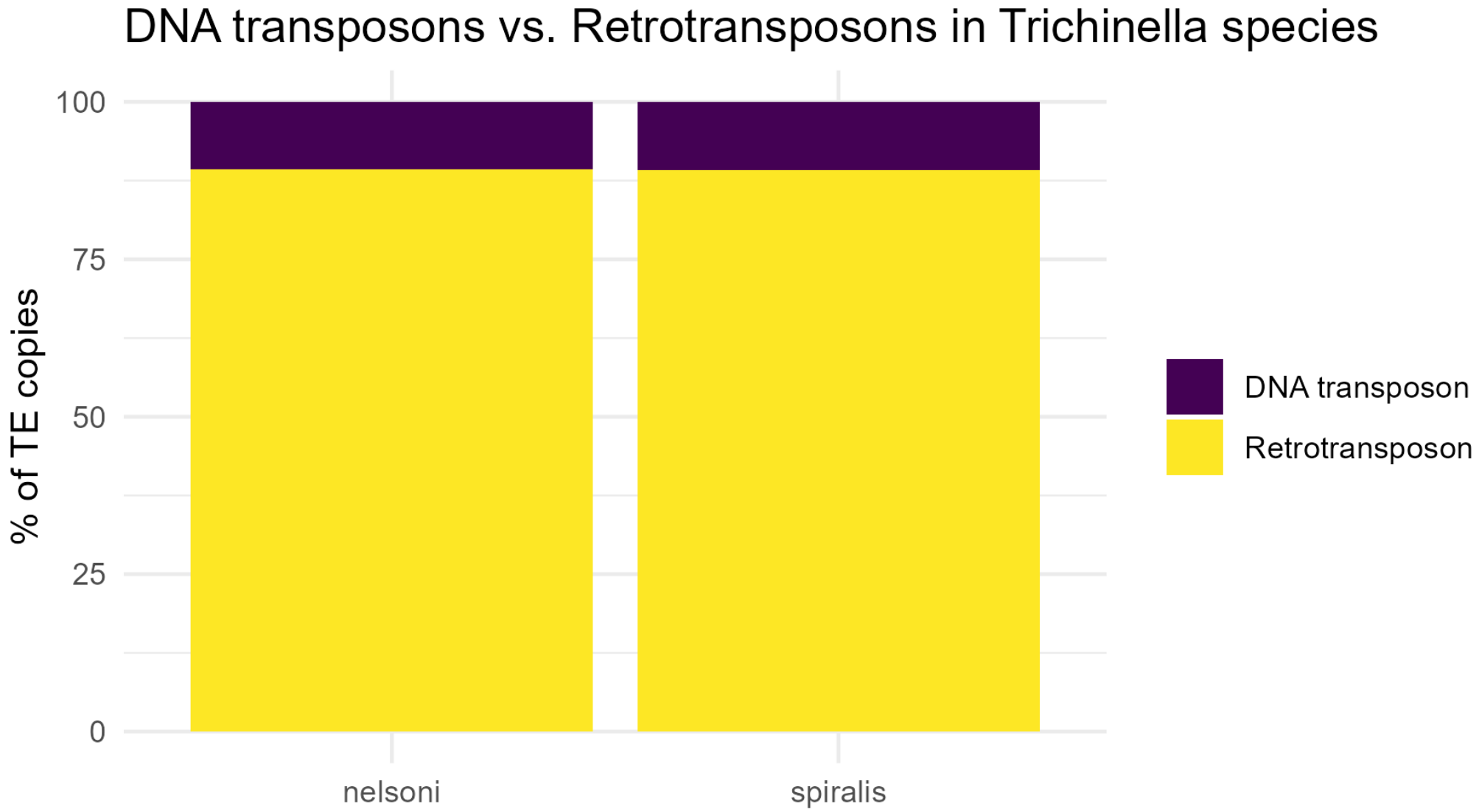


Figure 2. *T. spiralis* and *T. nelsoni* have highly similar retrotransposon superfamily compositions

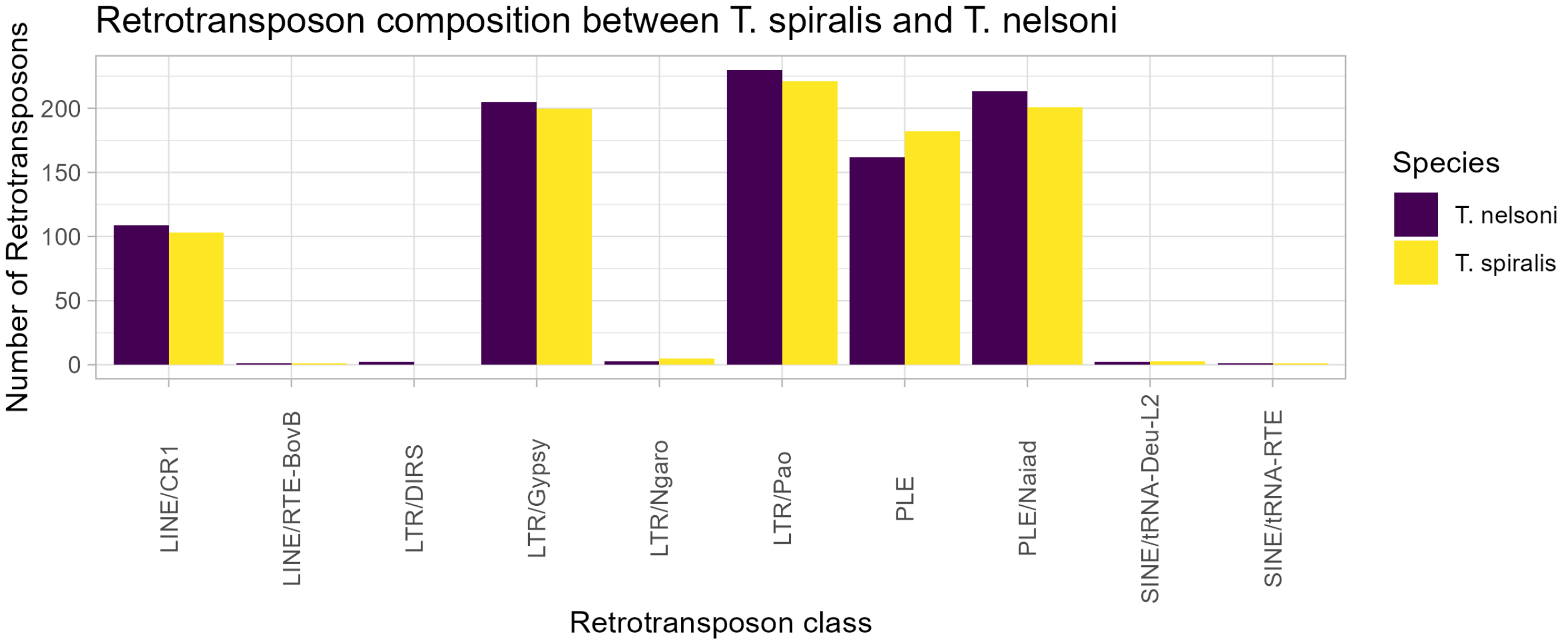
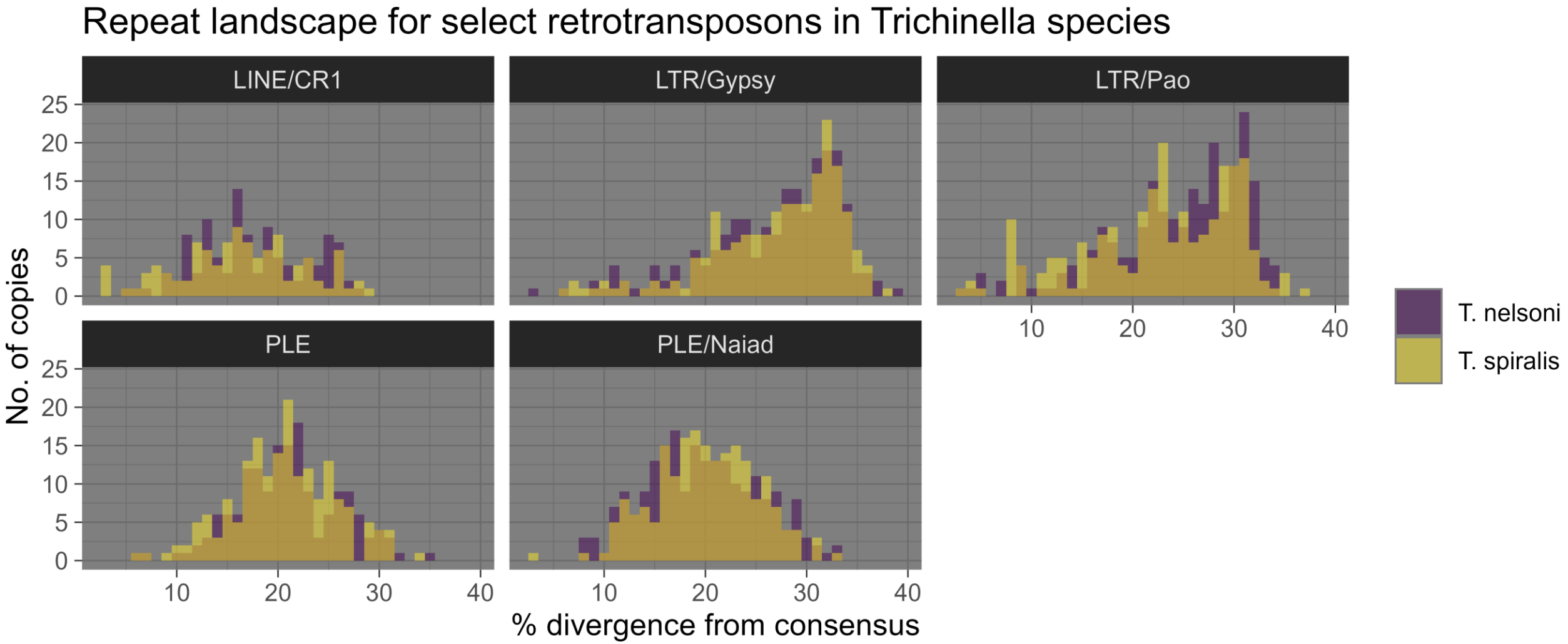


Figure 3. No statistical difference between the percent divergence of 5 selected retrotransposon superfamilies between *T. spiralis* and *T. nelsoni,* indicated by KS test (LINE/CR1 p = 1.00, LTR/Pao p = 0.209, PLE p = 1.00, PLE/Naiad p = 1.00, LTR/Gypsy p = 1.00)



Results/Discussion

Results: The figures reveal high similarity between the species

Figure 1 illustrates that the proportions of DNA transposons and retrotransposons are very similar in both species (chi-squared test: $X^2 = 0.25$, df = 1, p = 0.62).

Figure 2 shows similar retrotransposon counts in the same superfamilies in both species:

- Both have high counts of LTR/Gypsy, LTR/Pao, PLE and PLE/Naiad
- *T. nelsoni* is a little higher in LINE/CR1, LTR/Gypsy, LTR/Pao and PLE/Naiad
- *T. spiralis* is a little higher in PLE
- Caveat: did not statistically test differences in superfamily counts between the species.

Figure 3: Most retrotransposon copies are 10-30% diverged. Few are below 10%

- KS tests did not find a significant difference between the two species (adjusted p: LINE/CR1 = 1.00, LTR/Pao = 0.209, PLE = 1.00, PLE/Naiad = 1.00, LTR/Gypsy = 1.00).
- *T. spiralis* has a few copies with lower sequence divergence than 10% in LINE/CR1 and LTR/Pao (not statistically significant)

Discussion: Retrotransposon abundance does not correlate with level of infectivity

- We hypothesized that *T. spiralis* would have more retrotransposons and/or less diverged retrotransposons which might explain differences in infectivity capabilities of the two species.
- Our hypothesis was not supported because we did not find a significant difference between the two species when comparing DNA transposons vs. Retrotransposon counts or retrotransposon superfamily compositions.
- This may be because they are sister species, and probably inherited most TEs from a common ancestor (Korhonen *et al.*, 2016).
- Furthermore, most of the retrotransposon copies may be old, indicated by the relatively high percent divergence (Figure 3). The accumulated mutations likely reduces their activity, i.e. their ability to transpose in the genome.
- TE abundance does not necessarily reflect activity of the TEs currently.
- What we learned: TE amount alone is a weak predictor of infectivity. TE age, activity and where they are located near certain genes may be more important than TE counts to understand infectivity differences between species.
- **Next steps:** Since our methodology did not involve measuring TE activity or regulatory effects on nearby genes, future research could study TE insertion locations and their impact on expression of nearby genes. Also, comparing more *Trichinella* species could help provide a more substantiated look into whether certain TE superfamilies have specific biological differences. For example, comparing *Trichinella* from more distance relatives to parse how each class of TE impact species.

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Copilot Prompts

Copilot Prompts – Tomas Fischer

- "Find me scientific journal articles that compare different members of the trichinella genus, specifically including T. spiralis and T. nelsoni"
- Are there any that discuss transposible elements?
- Are there any that link transposible element classes to pathogenicity? What are the most important transposible elements in T. spiralis?
- with repeatmasker, what are the retrotransposons? everything with LTR and LINE?
- what about PLE's?
- what might be a good statistics test we can use to compare the differences between two repeatmasker data sets, specifically pertaining to retrotransposons in t. spiralis vs t. nelsoni?

Copilot Prompts – Bhumika Aparanji

- Which of the following "matching classes" identified in a RepeatMasker data set for Trichinella species are transposons and retrotransposons: "ARTEFACT", "DNA/hAT", "DNA/hAT-Ac", "DNA/MULE-MuDR", "DNA/PIF-Harbinger", "DNA/TcMar-ISRm11" , "DNA/TcMar-Mariner", "DNA/TcMar-Pogo", "DNA/TcMar-Tc1", "DNA/TcMar-Tc4", "LINE/CR1", "LINE/RTE-BovB", "Low_complexity", "LTR/Gypsy", "LTR/Ngaro", "LTR/Pao", "PLE", "PLE/Naiad", "rRNA", "Satellite", "Simple_repeat", "SINE/tRNA-Deu-L2", "SINE/tRNA-RTE", "snRNA", "Unknown"
- What is the biological and genetic relevance of percentage divergence?
- Which kind of statistical test is best suited to compare TE copies vs % divergence between two species for a subset of the matching classes?
- What is the Kolmogorov-Smirnov (KS) test?
- Why would a KS test give me approximate p-values?

Copilot Prompts – Adhithya Dhivakar

- Why study transposable elements (TEs) in *Trichinella*?
- Are *T. spiralis* and *T. nelsoni* sister species?
- Is *T. spiralis* more pathogenic than *T. nelsoni*?
- What paper supports the idea that *T. spiralis* is more infective than *T. nelsoni*?
- Is infectivity the same as host adaptation?
- Is *T. spiralis* a pathogen?
- What are the major classes of TEs?
- What is the difference between DNA transposons and retrotransposons?
- What is the difference between LTR and non-LTR retrotransposons?
- How to interpret TE abundance figures
- Whether two TE landscapes were significantly different
- How to compare TE composition between species
- Why were those retrotransposon families selected?
- Why are some families grouped together?
- How should I interpret the abundance patterns between *T. spiralis* and *T. nelsoni*?
- What is a KS test?
- What does "adjusted p-values" mean?
- Does adjusted p-value just mean multiple p-values?
- What are the KS tests comparing?
- What does it mean if all adjusted p-values are > 0.05?
- What does multiple-testing correction do?
- How do I interpret p = 0.6203?
- What does it mean when there is no statistically significant difference?
- How should I describe a chi-squared test result?
- How should I describe non-significant differences between species?
- What wording sounds more biologically accurate?
- How should I explain TE composition similarities?
- Can I say "According to the authors..."?
- Can I say "According to the paper..."?
- What are 3 key papers about transposable elements?
- What are 3 key papers about *Trichinella spiralis*?
- What are 2 key papers about TEs in *Trichinella*?
- Which papers support differences between *T. spiralis* and *T. nelsoni*?

Copilot Prompts – Purushoth Puvanendran

- "is this true: higher degree of retrotransposon content due to its increased pathogenicity? like in trichelle nelsoni?"
- "what mechanism would be responsible for increased pathogenicity?"
- "what is the relevance of percent divergence"
- "why would divergence indicate insertion? couldn't a mutated one also move? "
- "right but aren't transposons just sequences within a genome moving. they don't come from outside sources, so why would we expect less divergence being analog to fresh unmutated. "
- "so explain how retrotransposon in trinchelle have been implicated with pathogenecity, and are any specific of concern, such as LTR (Gypsy, Pao), LINE (CR1), SINE and PLE "
- "how do these two species vary: T. spiralis and T. nelsoni are closely related, but they have a significant difference in their pathogenicity , link this to studies."
- "more context on LINE, PLE, and DIRS as they all sound to do the same thing "

Copilot Prompts – Minaal Siddiqui

- Which one of these are retrotransposons? : 1 Simple_repeat 39421 2 Low_complexity 6728 3 rRNA 497 4 Unknown 404 5 LTR/Pao 323 6 PLE/Naiad 239 7 LTR/Gypsy 224 8 PLE 221 9 LINE/CR1 115 10 DNA/TcMar-Tc1 85 11 Satellite 57 12 DNA/PIF-Harbinger 27 13 DNA/MULE-MuDR 6 14 LTR/Ngaro 6 15 DNA/TcMar-Mariner 4 16 SINE/tRNA-Deu-L2 4 17 DNA/TcMar-Pogo 3 18 RC/Helitron 3 19 DNA/hAT-Ac 2 20 ARTEFACT 1 21 DNA/CMC-Chapaev 1 22 DNA/TcMar-Tc4 1 23 DNA/hAT 1 24 LINE/RTE-BovB 1 25 LTR/DIRS 1 26 SINE/tRNA-RTE 1 27 simple_repeat 1 28 snRNA 1

- What can LTR in Trichinella genus tell us?